

Lecture Notes for Neural Networks I



Data Quality and Imputation

Class Logistics and Agenda

- Logistics:
 - Need help? Use class GitHub as a resource!
 - TA hours coming soon!
 - Team Forming
 - Time to start Lab One
- Agenda:
 - Data Quality
 - Imputation methods
 - Data Representations

Class Overview, by topic

Table Data
Visualization

Numpy, Pandas, Seaborn
Overviews with some in-depth discussion

Linear and
Logistic
Regression

Numpy, Recreate API for Scikit-learn
Detailed mathematics for simple optimization
intuition for advanced optimization

Neural Networks
and Back Prop.

Numpy, Recreate API for DL Frameworks
Detailed mathematics for NN operations

Wide and Deep
Networks

Convolutional
Networks

Sequential
Networks

Keras, Tensorflow, Torch
Intuition, Detailed implement.

Large Language
Models (if time)

Hugging Face
Intuition only

Using the SMU HPC

- Tutorials available for various types of analysis:
 - <https://www.smu.edu/oit/research>

Events This Month

September 2026						
«						»
S	M	T	W	T	F	S
		1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30			
Upcoming Events:						
Data Science, AI/ML Office hour - PAB319B						
10:00am - 12:00pm, Wednesday, September 2, 2026						
Data Science workflow with R using caret						
1:00pm - 3:00pm, Wednesday, September 2, 2026						
Research Technology Office Hours						
10:00am - 12:00pm, Thursday, September 3, 2026						
Data Science, AI/ML Office hour - PAB319B						
10:00am - 12:00pm, Wednesday, September 9, 2026						
Data Science workflow with Python using scikit-learn						

Last Time

Data Quality Problems

- Missing
 - Easy to find, NaNs
- Duplicated
 - Easy to find, hard to verify
- Noise or Outlier
 - Hard to define
 - Hard to catch

Discrete	Nominal or Categorical	Variable could be one value in a set of categories. No ordering of values. <i>Example: Employee ID</i>	Allowed Transforms: permuting values boolean, one hot encoding, or hash function
	Ordinal	Variable could be one value in a set of categories. Ordering matters. <i>Example: Start Ratings, 1-5</i>	Allowed Transforms: $V_{new} = f_{mono}(V_{old}) + b$ integer (or boolean)
Continuous	Interval or Numeric	Value is continuous numeric value. Could be in specified range. <i>Example: BMI, Temperature, etc.</i>	Allowed Transforms: $V_{new} = f_{mono}(V_{old}) + b$ float
	Ratio or Numeric	Value is continuous numeric value. Zero is meaningful. Often not treated differently than interval. <i>Example: Length, Elevation</i>	Allowed Transforms: $V_{new} = f_{mono}(V_{old})$ float

Split-Impute-Combine

TID	Pregnant	BMI	Age	Diabetes
1	Y	33.6	31-40	positive
2	N	26.6	31-40	negative
3	Y	23.3	?	positive
4	N	28.1	21-30	negative
5	N	43.1	31-40	positive
6	Y	25.6	21-30	negative



split: pregnant
split: BMI > 32

TID	Pregnant	BMI	Age	Diabetes
1	Y	>32	31-40	positive
8	Y	>32	?	negative
10	Y	>32	21-30	positive

TID	Hair Color	Hgt.	Age	Arrested
1	Brown	5'2"	23	no
2	Hazel	1.5m	12	
3	NaN	5	999	no
4	Brown	5'2"	23	no

Self Test, Missing data

- Can all missing data be found by searching for NaNs?
 - A. Yes. Missing data should always be a NaN.
 - B. Yes. Pandas defaults all missing data to NaN.
 - C. No. This only works for floats, because that is the data type for NaN.
 - D. No. NaN only represents missing data that is already found.

Split-Impute-Combine



<i>TID</i>	<i>Pregnant</i>	<i>BMI</i>	<i>Age</i>	<i>Diabetes</i>
1	Y	33.6	31-40	positive
2	N	26.6	31-40	negative
3	Y	23.3	?	positive
4	N	28.1	21-30	negative
5	N	43.1	31-40	positive
6	Y	25.6	21-30	negative
7	Y	31.0	21-30	positive
8	Y	35.3	?	negative
9	N	30.5	51-60	positive
10	Y	37.6	21-30	positive

split: pregnant
split: BMI > 32

<i>TID</i>	<i>Pregnant</i>	<i>BMI</i>	<i>Age</i>	<i>Diabetes</i>
1	Y	>32	31-40	positive
8	Y	>32	?	negative
10	Y	>32	21-30	positive

Mode: none, can't impute

<i>TID</i>	<i>Pregnant</i>	<i>BMI</i>	<i>Age</i>	<i>Diabetes</i>
3	Y	<32	?	positive
6	Y	<32	21-30	negative
7	Y	<32	21-30	positive

Mode: 21-30

Idea: Imputing simple stats in smaller groups is more relevant.

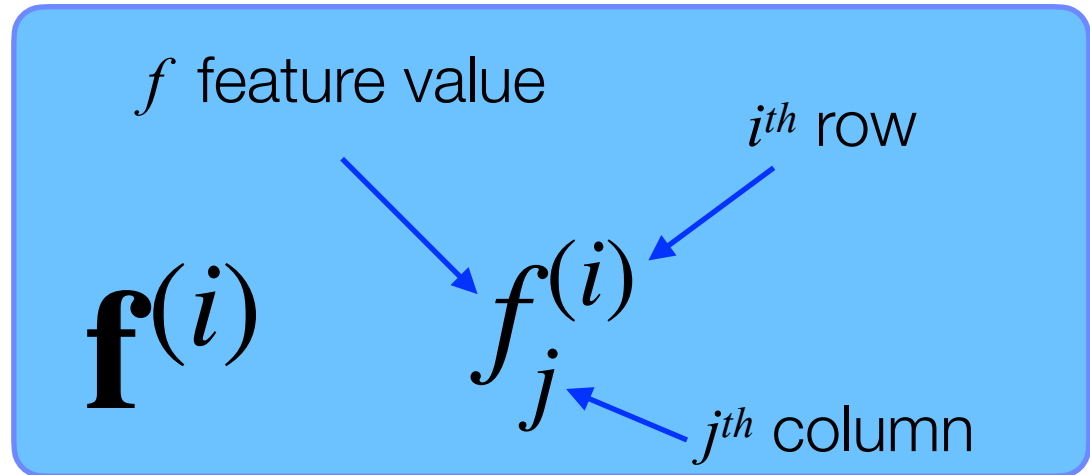
Group similar rows together first,
then **impute within the group**

Distance between Observations

Best for ML
Applications

	1	2	3	4
<i>TID</i>	<i>Pregnant</i>	<i>BMI</i>	<i>Age</i>	<i>Diabetes</i>
1	Y	33.6	31-40	positive
2	N	26.6	31-40	negative
3	Y	23.3	?	positive
4	?	28.1	21-30	negative
5	N	43.1	31-40	positive
6	Y	25.6	21-30	negative
7	Y	31.0	21-30	positive
8	Y	35.3	?	negative
9	N	30.5	51-60	positive
10	Y	37.6	21-30	positive

Some notation for the course:



$f_4^{(5)} = \text{positive}$	$\mathbf{f}^{(2)} = [\text{N}, \dots, \text{neg}]$
$f_2^{(10)} = 37.6$	$f_3^{(8)} = \text{NaN}$

Distance between
 unk row and i^{th} row

$$d^{(i)} = \frac{1}{|F_{valid}|} \sum_{j \in F_{valid}} \|f_j^{(unk)} - f_j^{(i)}\|$$

K-Nearest Neighbors Imputation

Best for ML Applications

TID	Pregnant	BMI	Age	Diabetes
1	Y	33.6	31-40	positive
2	N	26.6	31-40	negative
3	Y	23.3	?	positive
4	?	28.1	21-30	negative
5	N	43.1	31-40	positive
6	Y	25.6	21-30	negative
7	Y	31.0	21-30	positive
8	Y	35.3	?	negative
9	N	30.5	51-60	positive
10	Y	37.6	21-30	positive

For $unk = 3$, find k closest neighbors

TID	Preg.	BMI	Age	Diabetes	Distance $d^{(i)}$
$unk=3$	Y	23.3	?	positive	0
$i=6$	Y	25.6	21-30	negative	$(0 + 2.3 + 1)/3$
$i=2$	N	26.6	31-40	negative	$(1 + 3.3 + 1)/3$
$i=4$	?	28.1	21-30	negative	$(4.8 + 1)/2$

... repeat for all rows, select closest ...

Imputed Age: 21-30

Distance can be calculated differently:

- Valid features only? Use Categorical?
- Normalize ranges?
- Weight neighbors differently?
- Use min # of valid features?
- Type: Euclidean, city-block, etc.

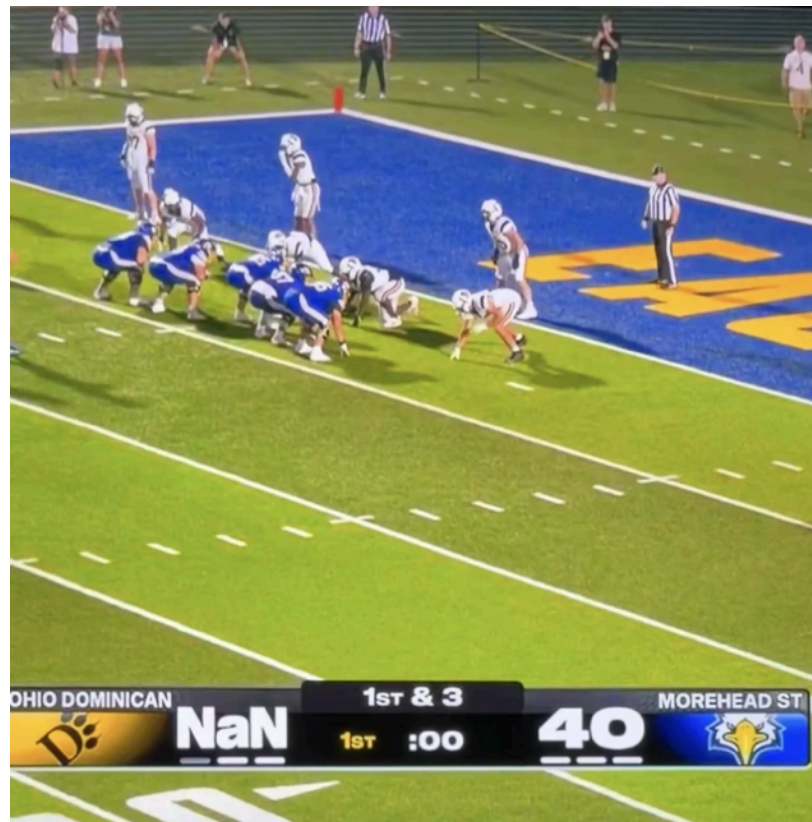
$$d^{(i)} = \frac{1}{|F_{valid}|} \sum_{j \in F_{valid}} \|f_j^{(unk)} - f_j^{(i)}\|$$

Pandas and Imputation
Scikit-Learn



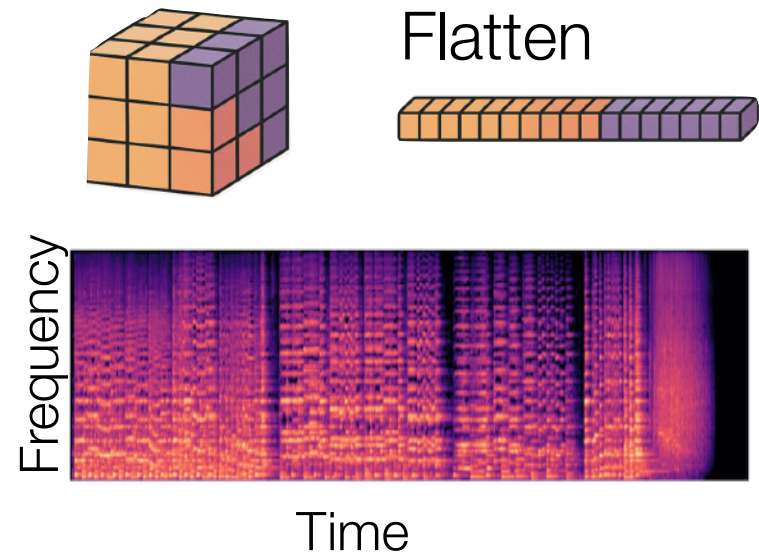
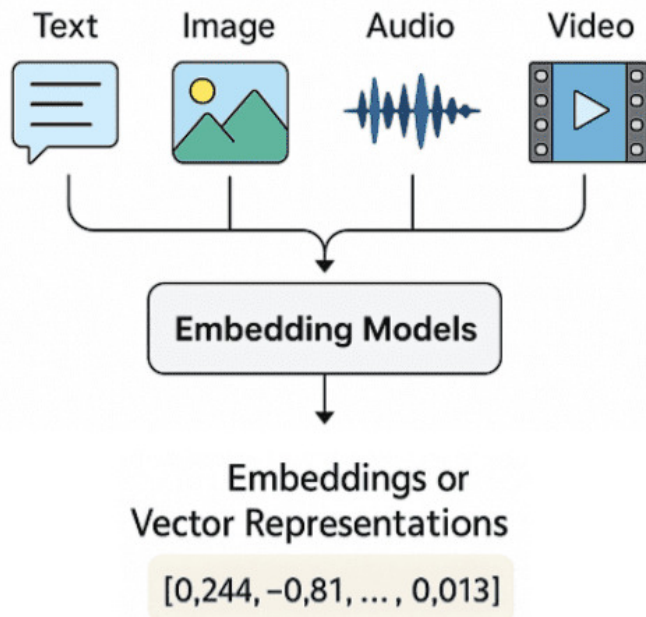
02. Data Visualization.ipynb

Data Representation and Documents



Common Data Encoding

- **Numeric:** Normalize and shift
- **Categorical:** One hot encode, then embed
- **Text** (large categorical): tokenize/embed, tf-idf, bag of words
- **Images/Video:** perhaps flatten and normalize, process in 2D windows, patches and embed
- **Time series:** normalize and process in windows



Data Tables as Variable Representations

if
time

Table

<i>TID</i>	<i>Pregnant</i>	<i>BMI</i>	<i>Age</i>	<i>Eye Color</i>	<i>Diabetes</i>
1	Y	33.6	41-50	brown	positive
2	N	26.6	31-40	hazel	negative
3	Y	23.3	31-40	blue	positive
4	N	28.1	21-30	brown	inconclusive
5	N	43.1	31-40	blue	positive
6	Y	25.6	21-30	hazel	negative

Internal Rep.

<i>TID</i>
1
2
3
4
5
6

Data Tables as Variable Representations

if
time

Table

<i>TID</i>	<i>Pregnant</i>	<i>BMI</i>	<i>Age</i>	<i>Eye Color</i>	<i>Diabetes</i>
1	Y	33.6	41-50	brown	positive
2	N	26.6	31-40	hazel	negative
3	Y	23.3	31-40	blue	positive
4	N	28.1	21-30	brown	inconclusive
5	N	43.1	31-40	blue	positive
6	Y	25.6	21-30	hazel	negative

Internal Rep.

<i>TID</i>	<i>Binary</i>	<i>Float</i>	<i>Ordinal</i>	<i>Object</i>	<i>Diabetes</i>
1	1	33.6	2	hash(0)	1
2	0	26.6	1	hash(1)	0
3	1	23.3	1	hash(2)	1
4	0	28.1	0	hash(0)	2
5	0	43.1	1	hash(2)	1
6	1	25.6	0	hash(1)	0

Bag of words model

if
time

<i>TID</i>	<i>Pregnant</i>	<i>BMI</i>	<i>Chart Notes</i>	<i>Diabetes</i>
1	Y	33.6	Complaints of fatigue wh...	positive
2	N	26.6	Sleeplessness and some...	negative
3	Y	23.3	First saw signs of rash o...	positive
4	N	28.1	Came in to see Dr. Steve...	inconclusive
5	N	43.1	First diagnosis for hospit...	positive
6	Y	25.6	N/A	negative

Bag of Words

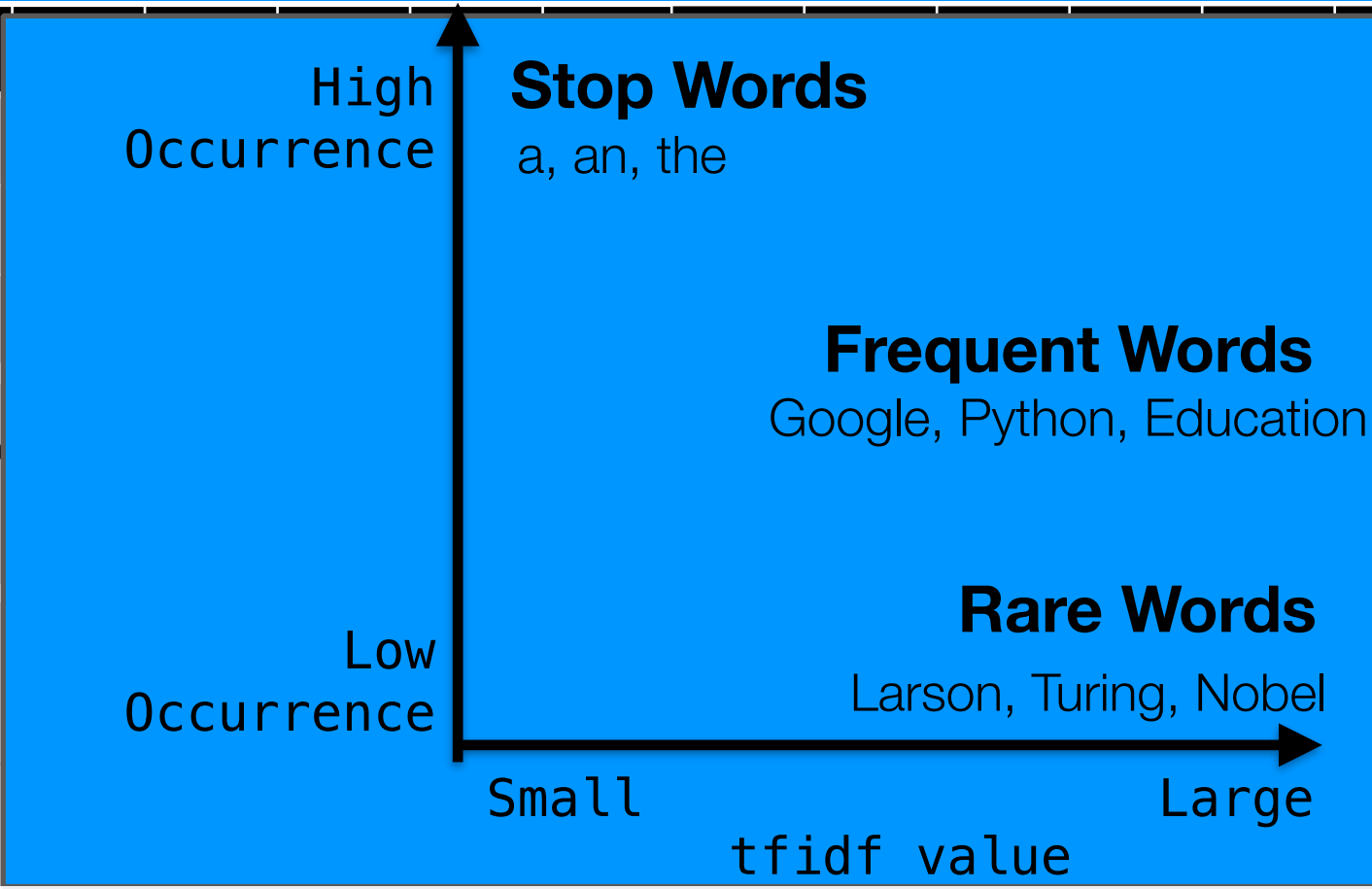
Vocabulary						
TID	Sleep	Fatigue	Weight	Rash	First	Sight
1	0	1	0	0	2	0
2	1	1	0	0	1	1
3	1	1	0	2	1	1

number of occurrences

Term-Frequency, Inverse-Document-Frequency

TID	Slee
1	0
2	0.1
3	0.1

tc	Stev
.86	0
.02	0
.1	0



This is often used in RAG systems, for Keyword retrieval!!

Want to know more?

Take Natural Language Processing!

For Next Lecture

- Before next class:
 - verify installation of seaborn, plotly, (and/or bokeh if you want)
 - look at pandas table data and additional tutorials
- Next time: Data Visualization

Lecture Notes for **Neural Networks I**

Professor Eric Larson
Data Quality and Imputation